MERN STACK INTERN

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Task Report: (Advanced CSS)

Project: Responsive Feature

Module: Week 2, Day 2 – Forms, Accessibility (a11y), and UI Polish

Focus: Semantic HTML, Accessibility (ARIA), HTML Form Validation, and Advanced CSS Styling

GitHub Link ***https://github.com/955muneeb/Glaxit-internship/tree/main/week-2/day-2***

1. Objective

The objective of this task was to design and develop a modern, accessible, and user-friendly client lead intake form using **HTML5** and **CSS3**. Rather than relying on the browser's default form appearance, the focus was on creating a fully customized interface that provides clear visual feedback, follows accessibility best practices, and delivers an improved user experience without using JavaScript.

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2. HTML5 Form Validation Using CSS (:valid and :invalid)

Modern browsers include built-in form validation that automatically verifies user input based on HTML attributes such as:

- required
- type="email"
- type="tel"
- minlength
- maxlength
- pattern

Instead of writing JavaScript for basic validation, CSS pseudo-classes were used to visually represent the validation state of each input field.

Valid State (:valid)

When a user enters data that satisfies all validation rules, the browser marks the input as **valid**.

For example:

- A correctly formatted email address such as **name@company.com**
- A required field that has been filled correctly

When this state is detected, CSS automatically applies:

- A green left border
- A subtle light-green background

These visual indicators immediately inform users that their input has been accepted successfully.

Invalid State (:invalid)

If the entered data does not satisfy the validation requirements, the browser marks the input as **invalid**.

Examples include:

- An incorrectly formatted email address
- A required field left empty
- Input that does not match the specified pattern

The invalid state is highlighted using:

- A red left border
- A soft red background

This provides instant feedback, allowing users to identify and correct mistakes before submitting the form.

Preventing Validation on Initial Page Load

A common usability issue occurs because empty required fields are technically considered invalid as soon as the page loads.

To avoid displaying error styles before the user interacts with the form, the following selector was implemented:

input:not(:placeholder-shown):valid

input:not(:placeholder-shown):invalid

The :not(:placeholder-shown) pseudo-class ensures that validation styles are only applied **after the user begins entering data**.

This results in a cleaner interface by preventing unnecessary error messages or red highlights when the form first opens.

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3. Accessibility (a11y) Implementation

Accessibility was treated as an important part of the development process to ensure that users who rely on assistive technologies, such as screen readers, can interact with the form effectively.

Semantic Form Structure

The form fields were grouped using <fieldset> and <legend> elements.

This provides meaningful structure for assistive technologies, allowing screen readers to identify related inputs as a single logical section rather than independent fields.

ARIA Attributes

The following ARIA attributes were implemented to improve accessibility:

aria-required="true"

This attribute informs screen readers that a field is mandatory.

Unlike visual indicators such as a red asterisk (*), ARIA allows visually impaired users to receive the same information through spoken feedback.

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4. User Interface Enhancements

To improve usability and create a more interactive experience, several visual enhancements were applied.

Custom Focus State

The browser's default focus outline was replaced with a customized focus style that includes:

- A bright orange border
- A pronounced drop shadow
- A slight upward movement (2px transform)

These effects make the currently active input field immediately recognizable, improving navigation and enhancing the overall user experience.

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5. Key Features Implemented

- Semantic HTML5 form structure
- Built-in HTML5 form validation
- Real-time CSS validation using :valid and :invalid
- Validation delayed using :not(:placeholder-shown)
- Accessible form design following ARIA best practices
- Screen reader support using aria-required and aria-describedby
- Custom focus animations
- Responsive and modern UI design
- Pure HTML and CSS implementation without JavaScript

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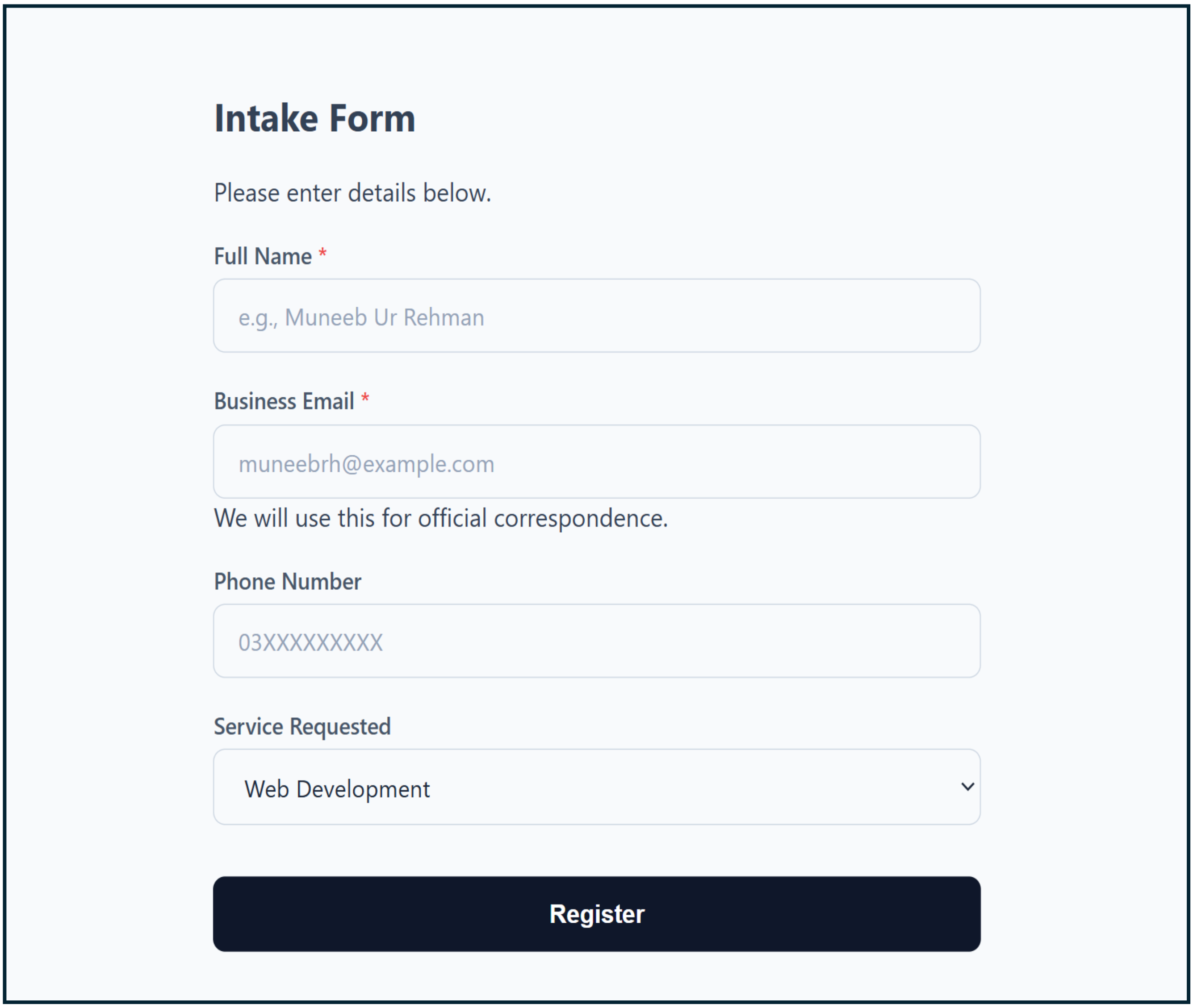
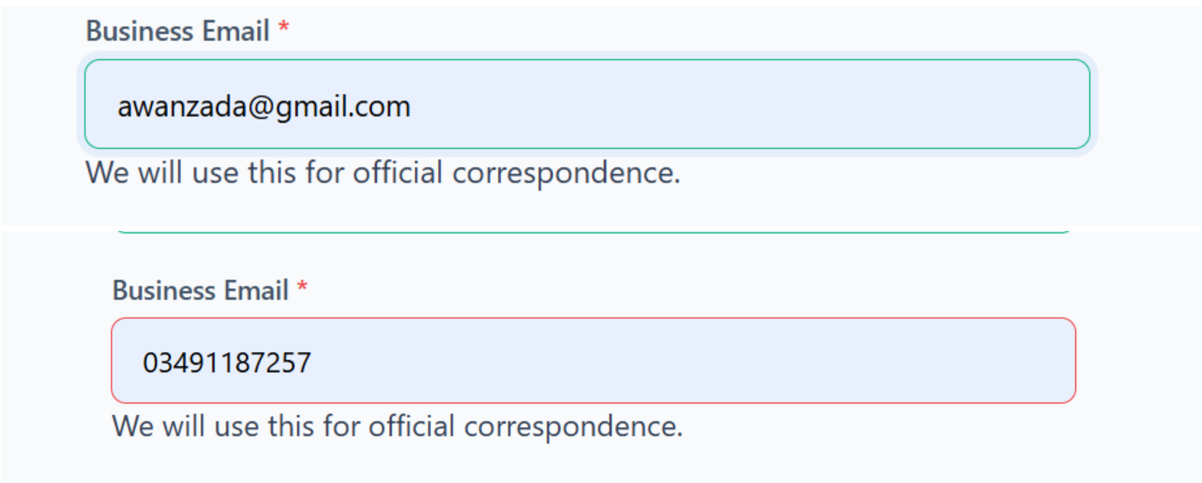
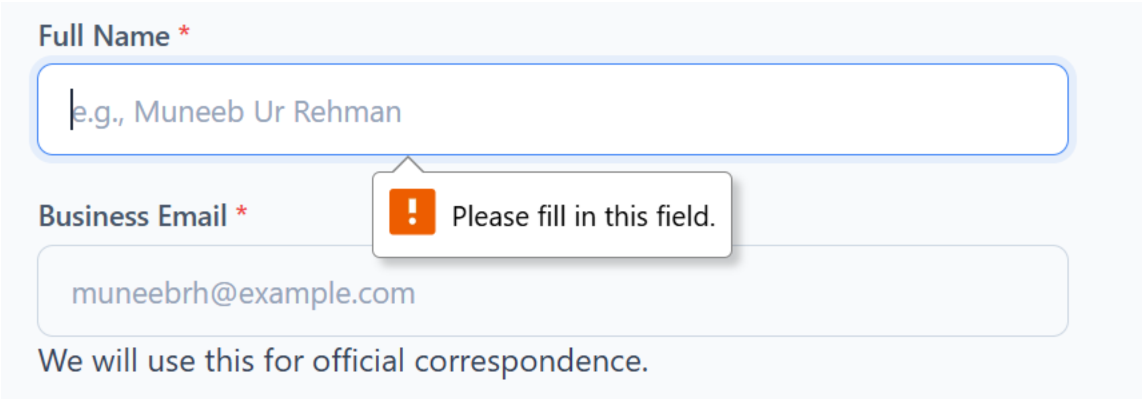
6. Output

Screenshot of the completed Lead Intake Form







7. GitHub Repository

Project Source Code:

https://github.com/955muneeb/Glaxit-internship/tree/main/week-2/day-2